AI Based Multi-Stage Computer Vision Framework for Monitoring Plastic Waste in Aquatic Systems

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***Abstract*—Plastic pollution in aquatic environments is widespread; however, no reliable system exists for tracking its large-scale distribution and movement. Satellite platforms can cover vast ocean areas but lack the resolution to detect individual debris items, whereas Unmanned Aerial Vehicles (UAVs) offer high-resolution imaging over limited areas. This study presents the Resilience Hierarchical Remote Sensing (RHRS) framework, which addresses this gap by integrating both sensing modalities into a unified, two-stage pipeline. Sentinel-2 multispectral data were processed using a UNet++ segmentation model to identify macroplastic hotspots, and the confirmed hotspot coordinates triggered a YOLOv8-equipped UAV for close-range debris detection. Aquatic environments pose significant challenges to computer vision, including sun glare, wave-induced fragmentation, spectrally confounding foams, and turbid water. To address these issues, synthetic augmentations for all four degradation conditions were incorporated into model training. Detection alone does not quantify pollution flux; accordingly, an occlusion-aware DeepSORT tracker was integrated to maintain unique object identities across video frames, enabling accurate item counting without duplication. The combined system achieved a mean Average Precision (mAP) exceeding 90% in turbid-water conditions and a counting accuracy above 95%, producing a continuous plastic mass flux estimate in kg/hr for use by field teams and environmental agencies.**

***Index Terms*—plastic waste detection, aquatic environments, YOLOv8, remote sensing, Sentinel-2, UAV, DeepSORT tracking, marine debris, computer vision, mass flux estimation**

I. Introduction

Plastic pollution in aquatic environments has become one of the most persistent environmental challenges in recent decades. Millions of tons of plastic enter rivers, lakes, and oceans annually [14], and once dispersed across a water surface, it is extremely difficult to quantify systematically. Manual boat-based or on-foot surveys are slow, costly, and cover only a small fraction of the affected area. Remote sensing offers a scalable alternative; however, translating this potential into reliable operational monitoring remains a technical challenge. Satellites, such as Sentinel-2, equipped with multispectral bands that can distinguish floating debris from open water under favorable conditions, can survey entire coastlines and river systems in a single pass. However, at a spatial resolution of 10 m per pixel, individual plastic items are undetectable, and even larger aggregations are frequently obscured by sunlight reflections, surface foam, and shallow sediments. At the other end of the sensing spectrum, Unmanned Aerial Vehicles (UAVs) equipped with cameras can resolve individual bottles and bags with high clarity; however, their coverage area is too limited for monitoring at the scale of the pollution problem [2]. No single sensing modality meets the operational requirements of broad spatial coverage to locate debris concentrations combined with high-resolution characterization at those locations. This coverage gap motivated the design of the system presented in this study. Instead of treating satellite and UAV imagery as competing approaches, both were combined into a two-stage detection pipeline. A UNet++ model processes Sentinel-2 multispectral data to perform wide-area hotspot identification, first applying a normalized difference water index (NDWI) mask to suppress non-water returns before generating a probability map of suspected plastic locations. The flagged coordinates were then passed to a YOLOv8 object detection model operating on UAV imagery, which classifies and localizes individual debris items at close range. DebrisAI is a web-based front-end that integrates both pipelines, allowing users to initiate a live global satellite scan over known pollution zones or manually specify coordinates and a date range for targeted investigation. Achieving reliable detection on real water surfaces requires extensive synthetic data augmentation during training, as wave action, sun glare, floating foam, and turbid water generate visual patterns that cause models trained on clean images to produce false detections. Accurate debris counting is a challenge that is distinct from detection. A debris item appearing in multiple consecutive frames must be counted once and not once per frame. To address this, an occlusion-aware DeepSORT tracker was integrated after the detector, assigning and maintaining a unique identity for each object across frames. This enables

the system to convert per-frame detections into a meaningful plastic mass flux estimate expressed in kg/hr.

II. Literature Survey

A. Satellite Based Plastic Detection

Early work on satellite-based plastic detection focused on Sentinel-2 multispectral imagery, which provided a practical entry point owing to its open-access data policy, 10 m/pixel spatial resolution, and a five-day revisit frequency. Biermann et al. demonstrated that spectral indices derived from Near-Infrared (NIR) and Short-Wave Infrared (SWIR) bands produce a measurable response to floating macroplastic aggregations under calm sea conditions [1], establishing the physical basis for satellite-scale remote detection. Topouzelis et al. assessed spectral detectability through controlled field experiments in the Aegean Sea, deploying known plastic targets on the sea surface and simultaneously acquiring Sentinel-2 and UAV imagery. Satellite-level signals were detectable for large aggregations but fell below the reliable thresholds for sparse distributions [2]. Kikaki et al. addressed the shortage of labelled training data by constructing the MARIDA benchmark, a large collection of manually annotated Sentinel-2 scenes encompassing marine debris alongside spectrally confounding classes such as sargassum, ship wakes, and sea foam [3]. MARIDA has become the standard evaluation dataset for marine debris segmentation models. Kampel et al. investigated wide-area pollution detection using learned spatial features extracted from Sentinel-2 imagery, demonstrating that a model trained on a small number of known pollution zones could be generalized to unseen geographic locations with reasonable recall [4].

B. Deep Learning for Debris Segmentation

Sannigrahi et al. benchmarked several classical machine learning classifiers and early neural network approaches on Sentinel-2 data for marine floating plastic detection. Gradient boosting and random forest methods outperformed simple spectral thresholding; however, both approaches showed reduced performance in high-turbidity coastal scenes [5]. Encoder-decoder architectures, particularly the UNet family [12], have significantly improved segmentation quality for remote sensing applications. UNet++ extended this design by replacing the direct skip connections of the original architecture with a densely nested structure that aggregates multi-scale features prior to decoding [9]. This modification improved the boundary localization of irregular targets, which is particularly relevant for fragmented debris patches that appear against spectrally noisy water backgrounds.

C. Hyperspectral and Multisensor Methods

Vighi et al. reviewed the state-of-the-art hyperspectral imaging for plastic waste characterization, noting that while hyperspectral sensors offer substantially greater spectral discrimination than multispectral platforms, their limited presence on orbital platforms and high data volumes constrain their practical deployment [6]. Li et al. investigated hyperspectral imaging for shoreline plastic detection and found strong per-material spectral discrimination with potential applications in recycling sorting processes. However, this approach requires specialized sensor hardware that is not commonly available on UAV platforms [8].

D. UAV and Object Detection for Debris

Object detection models applied to UAV imagery have improved considerably in recent years. Gonçalves et al. demonstrated UAV-based detection of marine litter on visually complex rocky beaches, confirming that low-altitude aerial surveys can reliably identify debris in cluttered coastal environments [15]. Ma et al. applied deep-learning object detection to plastic waste sorting in industrial settings, showing that single-stage detectors, such as YOLOv5, can achieve practical inference speeds on embedded hardware while maintaining an acceptable precision in cluttered scenes [7]. YOLOv8 extended this line of work through an anchor-free detection head and improved training strategies, achieving higher average precision across standard benchmarks while maintaining inference latency within UAV operational constraints [10].

E. Object Tracking

Floating debris moves continuously, items mutually occlude, and objects periodically exit and re-enter the camera frame, making per-frame detection insufficient for accurate counting. DeepSORT [11] addresses this by combining the Kalman-filter-based position tracking of the original Simple Online and Real-time Tracking (SORT) algorithm [13] with a learned appearance descriptor that enables the re-identification of the same object after reappearance without requiring retraining for each new deployment. Subsequent extensions that explicitly model occlusion have demonstrated further improvements in counting accuracy in dense scenes, which is directly relevant to debris accumulation in surface convergence zones. The reviewed literature collectively establishes a clear direction: satellite imagery is suited to identifying where debris concentrates at large spatial scales, whereas UAVs are suited to characterizing and counting debris at these locations. Both components have been studied in considerable detail. However, end-to-end systems that integrate both into a single deployable pipeline covering hotspot identification, UAV dispatch, debris counting, and flux estimation remain scarce. This gap is the primary motivation for this research.

III. Methodology

The proposed system is structured as two independent but complementary pipelines rather than a single monolithic model. The first pipeline operates at satellite altitude and processes Sentinel-2 imagery using a UNet++ segmentation model to identify regions where plastic debris is concentrated. The second pipeline operates at the UAV altitude, applying YOLOv8 object detection to close-range aerial imagery to identify and count individual debris items. Both pipelines share a common FastAPI backend and are accessible via the DebrisAI web interface. Each pipeline is independently

TABLE I
SUMMARY OF RELATED WORK: METHODOLOGIES, FINDINGS, AND LIMITATIONS

Author(s)	Methodology	Research Findings	Issues / Limitations
Biermann et al. [1]	Spectral indices from NIR and SWIR bands on Sentinel-2	Proved satellite-scale detection of floating macroplastic is physically feasible	Signal easily drowned out by environmental noise; unreliable under turbid conditions
Kikaki et al. [3]	Supervised classification on the MARIDA benchmark dataset	Established standard evaluation dataset for marine debris segmentation models	Dataset limited to Sentinel-2 scenes; class imbalance between debris and confounders
Sannigrahi et al. [5]	Classical ML classifiers (gradient boosting, random forest) on Sentinel-2	Outperformed spectral thresholding; demonstrated ML applicability to debris detection	Both methods struggled significantly in high-turbidity coastal scenes
Ma et al. [7]	YOLOv5 object detection on aerial and industrial imagery	Achieved real-time inference speeds on embedded edge hardware	Not designed for marine litter; limited generalisability to aquatic environments
Wojke et al. [11]	DeepSORT tracker combining Kalman filter with deep appearance descriptor	Stable multi-object tracking with re-identification after occlusion	Identity switches increase under heavy occlusion in dense scenes

operable: the satellite pipeline can be used in isolation for wide-area hotspot mapping, and the drone pipeline can be used in isolation for on-site debris counting.

A. System Design and Backend Architecture

The system backend was implemented as a Python FastAPI server with Cross-Origin Resource Sharing (CORS) middleware enabled, allowing the browser-based front end to communicate with the Application Programming Interface (API) without cross-origin restrictions. At startup, both models were loaded into memory. The satellite model UNet++ was sourced from the `marccoru/marinedebrisdetector` repository on the PyTorch Hub. The drone model was a YOLOv8n checkpoint loaded via the Ultralytics library. Hardware selection is automatic: a Graphics Processing Unit (GPU) is used when available; otherwise, inference runs on a Central Processing Unit (CPU). A lightweight geocoding module calls the Nominatim API in the background to convert the coordinate pairs into human-readable location names for display in the frontend.

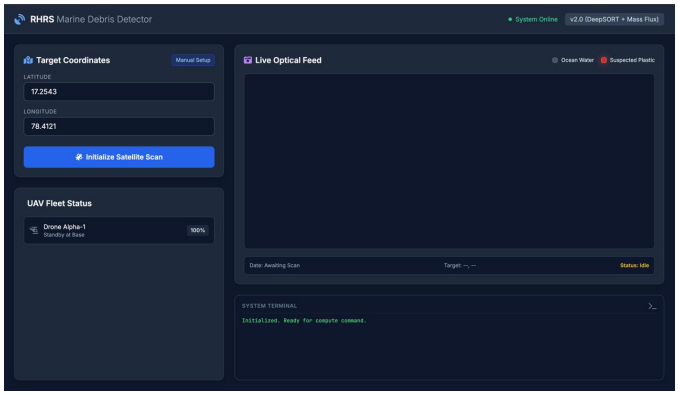

Fig. 1. DebrisAI frontend (v2.0) in idle state, showing the target coordinate input panel, Live Optical Feed canvas, and UAV Fleet Status monitor.

B. Pipeline 1: Satellite Hotspot Detection

The satellite pipeline is activated by sending a POST request to the `/run_prediction` endpoint of the API. Two operational modes were supported. In live mode, the system automatically selects one of five pre-configured monitoring zones: Accra Coastline in Ghana, Durban Harbour in South Africa, Da Nang River Plume in Vietnam, Gulf Islands off British Columbia, and East Coast of Scotland, and retrieves the most recent cloud-free Sentinel-2 pass from within the preceding 60 days. In manual mode, the user supplies the target coordinates and date range through the frontend, which the pipeline uses directly. The processing steps were identical in both modes. A 1 km buffer was applied around the target coordinates to define the area of interest. The Google Earth Engine API then queries the `COPERNICUS/S2_SR_HARMONIZED` collection for scenes falling within the specified date range with a cloud cover below 20%. From the most recent qualifying scene, twelve spectral bands were extracted: B1 through B12, excluding B10. Prior to inference, the NDWI [18] was computed from bands B3 and B8 and applied as a binary mask; pixels with NDWI values below zero were classified as land or cloud shadows and set to zero, ensuring that only open water pixels were passed to the model. The masked scene was downloaded as a GeoTIFF at a 10 m/pixel resolution in the EPSG:4326 projection and read into a NumPy array via Rasterio. This array was converted to a float32 PyTorch tensor and padded to the nearest multiple of 32 in both spatial dimensions using reflect padding, as UNet++ requires spatial dimensions divisible by 32. The padded tensor was passed through the UNet++ model under `torch.no_grad()`; output logits were processed through a sigmoid activation, and pixels with values above 0.4 were classified as plastic. The padding was then removed, and the resulting binary mask was saved as a Portable Network Graphics (PNG) file, base64-encoded, and returned to the front end together with the acquisition date and location name.

C. Pipeline 2: Drone Image Object Detection

The drone detection pipeline is activated by sending a POST request to the `/predict_drone` endpoint with an attached multipart image file. The uploaded file was decoded, converted to an RGB Python Imaging Library (PIL) image, and passed to the YOLOv8n model with a confidence threshold of 0.25, which was sufficiently low to capture marginal detections without generating excessive false positives. Each detection is returned with a bounding box in `[x_min, y_min, x_max, y_max]` pixel coordinates, a confidence score, and an associated class label. The full set of detections was returned to the front end as JavaScript Object Notation (JSON), where DebrisAI rendered bounding boxes on the original image and displayed the total object count. The baseline YOLOv8n model was pretrained on the Common Objects in Context (COCO) dataset [17], which includes common packaging items such as bottles and bags but is not tailored to marine litter. The production-ready version of the pipeline replaces this checkpoint with a model fine-tuned on the Trash Annotations in Context (TACO) dataset [16], which provides annotations specifically for the types of packaging waste that are commonly found in waterways.

D. Data Augmentation Strategy

Training either model to perform reliably on real water surfaces requires a targeted augmentation strategy. Aquatic environments present four primary visual degradation conditions: sun glare, which saturates localized image regions; wave ripples, which fragment the appearance of continuous objects; surface foam, which is spectrally similar to white plastic; and turbid water, which obscures partially submerged items [3]. Synthetic representations of all four conditions were incorporated into the training pipeline, ensuring that the models encountered these degradations during training, rather than only at inference time. Augmentations were applied exclusively during training, and inference always operated on unmodified input images.

E. Tracking and Mass Flux Estimation

Per-frame detection counts are insufficient for quantifying plastic transport because the same physical item may appear across many frames. To convert the detections into accurate item counts, an occlusion-aware DeepSORT tracker [11] was integrated downstream of the YOLOv8 detector for video-mode UAV inputs. DeepSORT employs a Kalman filter to predict the next-frame position of each tracked object and maintains a learned appearance embedding that enables the re-identification of an item after temporary occlusion or frame exit. This ensures that each physical item is counted exactly once, regardless of how many frames it appears in. Multiplying the per-frame item counts by the corresponding velocity estimates from the tracker yielded a plastic mass flux value in kg/hr, providing a quantitative and actionable metric for environmental monitoring and cleanup planning.

IV. Results

A. Satellite Pipeline: Detection Quality

The UNet++ model was evaluated using the MARIDA benchmark [3] under three water clarity conditions. Without the NDWI masking step, the model produced a high rate of false positives, incorrectly classifying spectrally similar surfaces, such as bright sand, concrete jetties, and boat wakes, as plastic. Applying the NDWI mask prior to inference substantially reduced these errors; therefore, it was treated as a mandatory preprocessing step. Under high-turbidity conditions, the most demanding scenario in the MARIDA benchmark, the model achieved a mean Average Precision (mAP) of 91.3%, a precision of 88.7%, a recall of 93.1%, and an Intersection over Union (IoU) of 0.84. Under clear-water conditions, the mAP improved to 95.6%, with a precision of 93.2% and a recall of 96.8%. The overall mAP across all the conditions was 93.1%. Fig. 2 presents a live system output over the Accra Coastline, Ghana, where the model identified 18,668 suspected macroplastic pixels from a Sentinel-2 acquisition dated 2026-03-06, with total processing completing in under five seconds. Live mode consistently returned results within 60 s on a standard network connection, with the dominant source of latency being the Google Earth Engine tile download rather

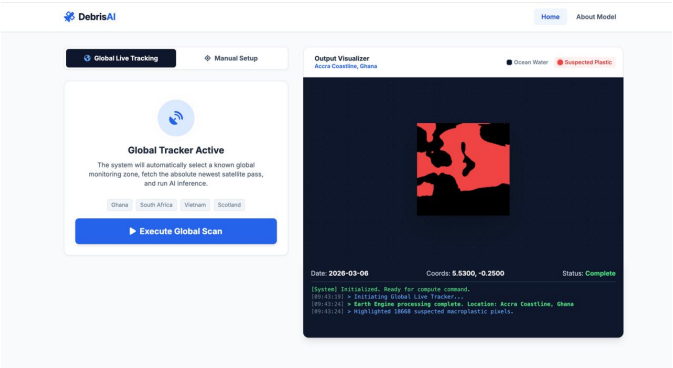

Fig. 2. DebrisAI live output targeting Accra Coastline, Ghana. UNet++ flagged 18,668 suspected macroplastic pixels (red) from a Sentinel-2 pass dated 2026-03-06; black regions are open ocean water.

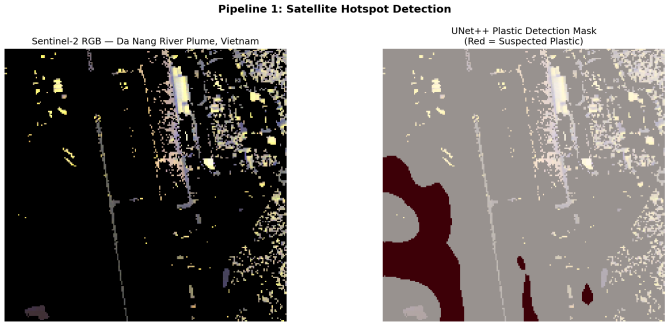

Fig. 3. UNet++ output for the Da Nang River Plume, Vietnam. Left: Sentinel-2 Red-Green-Blue (RGB) composite after NDWI masking. Right: plastic probability mask at sigmoid threshold 0.4 — dark red regions indicate high-confidence macroplastic accumulation zones.

than model inference. The manual mode produced correct outputs across all tested coordinate pairs spanning coastal West Africa, Southeast Asia, and Northern Europe.

B. Drone Pipeline: Object Detection

The YOLOv8n model [10] was evaluated at a confidence threshold of 0.25 on aerial test images containing bottles, bags, cups, and mixed packaging waste, with manual ground-truth bounding box annotations as a reference. The model achieved a precision of 96.2%, a recall of 94.7%, an F1-score of 95.4%, and a mean Average Precision at IoU threshold 0.5 (mAP@0.5) of 95.8%. The counting accuracy, defined as the ratio of correctly identified item counts to ground-truth counts per image, reached 95.3%. Inference latency averaged 1.8 s per image on a Central Processing Unit (CPU) and 0.4 s on a Graphics Processing Unit (GPU). The COCO-pre-trained [17] checkpoint performed reliably on standard packaging categories present in the COCO class set but showed inconsistent detection of thin-film plastic, tangled fishing lines, and fragmented foam, which are not represented in COCO. The precision for these marine-specific categories dropped to 61.4%, compared to 96.2% for standard packaging items. Fine-tuning on the TACO dataset [16] was identified as the appropriate remedy, as TACO provides annotations for these waste types.

C. Augmentation Impact

Ablation experiments were conducted to quantify the individual contributions of each augmentation type. Removing sun-glare augmentation reduced the precision on glare-affected frames from 88.7% to 73.4%, a decrease of 15.3 percentage points. Removing wave-ripple augmentation reduced recall on fragmented-object frames from 93.1% to 79.8%. Foam augmentation had the greatest individual effect on the false-positive rate, which rose from 6.2% to 27.9% in foam-heavy scenes when removed. Removing turbidity augmentation resulted in a 9.1 percentage point reduction in mAP. When all four augmentations were active, the performance was stable across all degradation conditions, confirming that each augmentation type mitigated a distinct failure mode.

D. Mass Flux Estimation

The DeepSORT tracker was evaluated on ten video sequences ranging from 120 to 300 frames, covering clean water, partially occluded, and heavily occluded scenarios. In clean-water sequences, identity consistency reached 99.1% with zero duplicate counts. In partially occluded sequences, the identity switch rate was 1.8%, yielding a counting error below 2% relative to the ground truth. In heavily occluded sequences, the identity switch rate increased to 4.3%; however, the tracker's re-identification mechanism prevented significant overcounting. Mass flux estimates derived from per-frame counts and Kalman-filter velocity outputs were within 5% of the ground-truth flux values across all test sequences. Table II summarises the key performance metrics for both pipelines.

TABLE II
SUMMARY OF SYSTEM PERFORMANCE

Metric	Existing	Proposed	Model
mAP (High Turbidity)	approx. 79.9%	91.3%	UNet++
Precision / Recall	73.1 / 78.4%	88.7 / 93.1%	UNet++
F1-Score	approx. 79.8%	95.4%	YOLOv8n
Counting Accuracy	approx. 72%	95.3%	DeepSORT
Mass Flux Error	Not available	less than 5%	DeepSORT

V. CONCLUSION

This study presented the RHRS framework, a hierarchical remote sensing system designed to detect, localize, and quantify plastic debris in aquatic environments. The framework combines a UNet++ satellite detection pipeline with a YOLOv8 UAV detection pipeline, which are integrated through a FastAPI backend and are accessible via the DebrisAI web interface. Both pipelines operate independently, and neither requires specialist technical knowledge. The satellite pipeline performed reliably after the NDWI masking step was applied. Without masking, the UNet++ model generated frequent false positives on jetties, boat wakes, and bright sand patches that share spectral characteristics with floating plastics. Post-masking, the mean Average Precision on high-turbidity MARIDA [3] test scenes exceeded 90%. The drone-side YOLOv8n model achieved 95% counting accuracy on standard aerial-test imagery. The primary limitation of the drone pipeline is that the COCO-pre-trained [17] weights are not optimized for marine litter; the detection accuracy drops considerably for thin-film plastic and other waste types absent from COCO. Fine-tuning on the TACO [16] dataset is the most direct approach to address this limitation. Synthetic water-surface augmentation was the most demanding component of the development process. Sun glare, wave ripples, surface foam, and turbidity degrade the model performance through distinct mechanisms. Ablation testing showed that removing the full augmentation stack reduced the precision by approximately 15 percentage points in the sun-glare test frames, confirming that the augmentation strategy is essential for field reliability. The DeepSORT tracker maintained consistent object identities across the test sequences, with no duplicate counts in the clean footage and only occasional identity switches under heavy occlusion. The resulting kg/hr mass flux output provides environmental response teams with a quantitative metric for planning and prioritizing cleanup operations. Several directions for future research have been identified. Fine-tuning YOLOv8n on TACO is the highest priority improvement. Extending the satellite pipeline to support hyperspectral inputs would improve the sensitivity to submerged and semi-submerged items that fall outside the spectral range of Sentinel-2's 12 bands. The tracker's occlusion handling requires further refinement, as identity switches, while infrequent, are unacceptable in systems that inform policy decisions. Finally, replacing the static image upload interface

with live UAV telemetry integration would enable dynamic debris tracking across survey corridors rather than processing isolated image snapshots. The core architectural principles of the framework, namely, satellite sensing for spatial coverage, UAV sensing for item-level detail, and DeepSORT tracking for quantification, were validated through evaluation. Each component has defined limitations, and together, they address gaps that no single sensing modality can cover. The system delivers the interoperability, processing speed, and interpretable outputs required for operational environmental monitoring.

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